

		2
22	A	Effective resistance is made up of copper and iron resistors in parallel. Per unit length, $\text{Copper resistance} = \frac{1.7 \times 10^{-8}}{2.0 \times 10^{-3}} = 8.5 \times 10^{-6} \Omega$ $\text{Iron resistance} = \frac{1.0 \times 10^{-7}}{1.0 \times 10^{-3}} = 1.0 \times 10^{-4} \Omega$ Effective resistance per unit length $= \left(\frac{1}{8.5 \times 10^{-6}} + \frac{1}{1.0 \times 10^{-4}} \right)^{-1} = 7.8 \times 10^{-6} \Omega \text{ m}^{-1}$
23	C	The smallest reading on the ammeter means smallest current passing through it